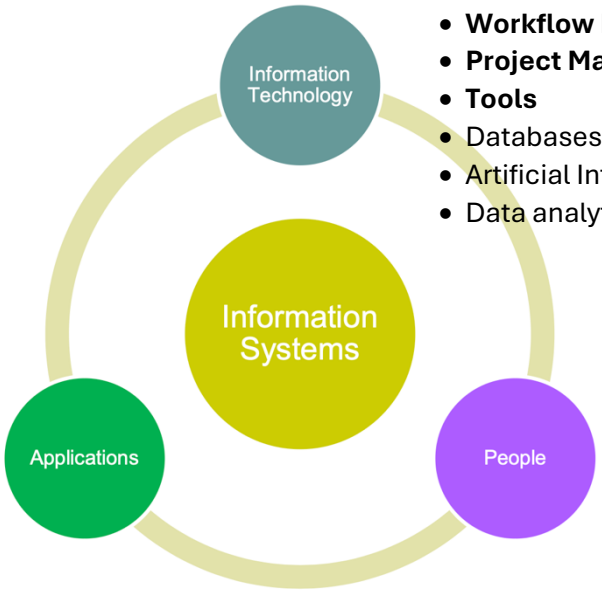
Lecture 1

❖ Business Informatics Overview



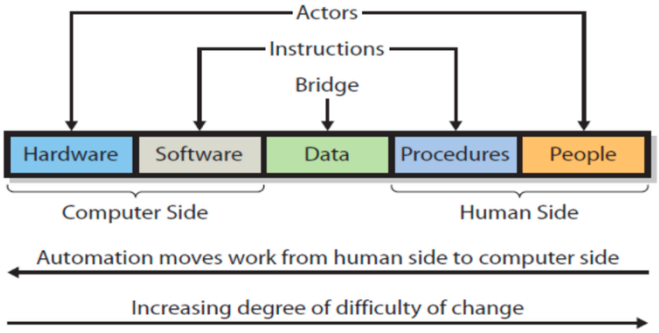
- **Workflow Management**
- **Project Management**
- **Tools**
- Databases
- Artificial Intelligence
- Data analytics

- **Social** aspects / effects
- **Technology** acceptance
- **IT (projects) management**
- **Information management**
- **Policies**, laws and privacy

- E-Business / E-gov
- **Enterprise systems implementation**
- **S/W development process management**
- Business Intelligence
- Data science (stats)

❖ Information system

Definition: Set of **interrelated components** collect, process, store, and distribute information



❖ Project

➢ Definition

- a **temporary endeavor** undertaken to accomplish a **unique product or service**
- **one-time processes, complex** create a unique product/service, **limited** by budget, schedule, and resources, **developed** to create business value

➢ Characteristics/Attributes

1. **unique** purpose
2. some **organizational structure**
3. **temporary**
4. **require resources**, often from various areas
5. should have a **primary sponsor** and/or **customer**
6. **involve uncertainty**

Process vs Project

Process	Project
Repeat process or product	New process or product
Several objectives	One objective
Ongoing	One-shot-limited life
People are homogenous	More heterogeneous
Well-established systems	Integrated system efforts
Greater certainty	Greater uncertainty
Part of line organization	Outside of line organization
Established practices	Violates established practice
Supports status quo	Upsets status quo

➢ **Stakeholders:** people involved in or affected by project

Examples: **project sponsor** and **project team,** support **staff, customers, users, suppliers, opponents** to the project

- **Parts of Project**

  - **IT Projects: (information technology)** involving **hardware, software, and networks** (How do we **build/run the system**?)
  - **IS Projects: (information system)** involving **applications** that serve the **informational needs** of IS users (How do **people use the system**?)
  - **SD Projects: (system development) :** refers to **projects producing new IS** that did not exist before
  - **Example**: **Setting up servers**, internet, and security (IT part) and **Building the website** where students watch lectures and submit assignments (IS Part)

- **Project Success and Failure**

  - **Success:** a project that delivers to the customer everything specified, to the quality agreed, **on time and within costs**
  - **Challenged:** the project was **completed and became operational** but **cost more**, **overran on time** and **delivered less functionality** or bad quality
  - **Impaired:** cancelled during the **development stage**

- **Faliure Factors**

  - **Runaway projects:** Exceed schedule, budget + fails to perform as specified
  - **System Failure:** Fail to capture essential business requirements + Fail to provide organizational benefits

❖ The Triple Constraint

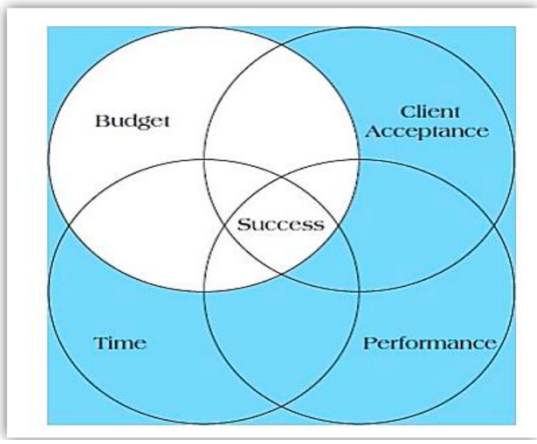
1. **Scope goals:** What is the **project trying to accomplish**?
2. **Time goals**: **How long should it take** to complete?
3. **Cost goals**: What should it **cost**?

❖ Project success

➢ *Quadruple constraint*

➢ *<u>Criteria for success</u>*

1. System quality
2. Info quality
3. Use
4. User satisfaction
5. Individual impact
6. Organizational impact



❖ Project Manager

➢ Role

- Selecting team
- Developing project objectives
- Risk management
- Cost estimation
- Scheduling
- Managing resourses

❖ Project Management

➢ **Definition:** is the **application** of knowledge, skills, tools, and **techniques** to **project activities** to achieve **specific targets within specified budget and time** constraints.

➢ **Life cycle:**

- Initiation -> Planning -> Execution -> Monitoring/Controlling -> Closure

➢ **Major variables:**

- Scope, time, cost, quality, risk

➢ **Knowledge areas (Areas):**

- 4 **core knowledge** areas lead to specific project objectives (scope, time, cost, and quality)
- 4 **facilitating knowledge** areas the means through which the project objectives are achieved (human resources, communication, risk, and procurement management)
- 1 **knowledge** area affects and is affected by all the other knowledge areas (project integration management**)**

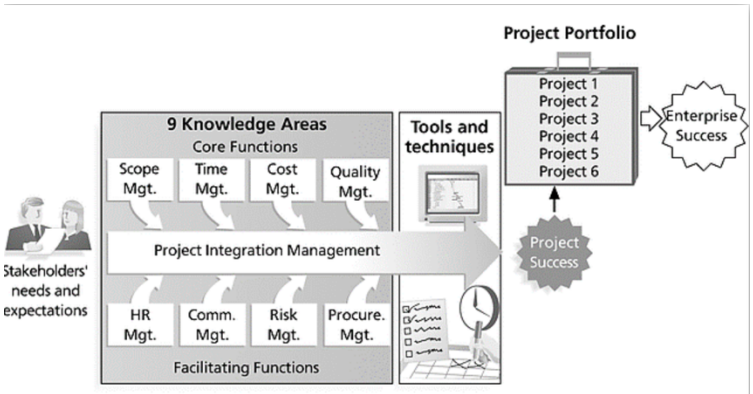
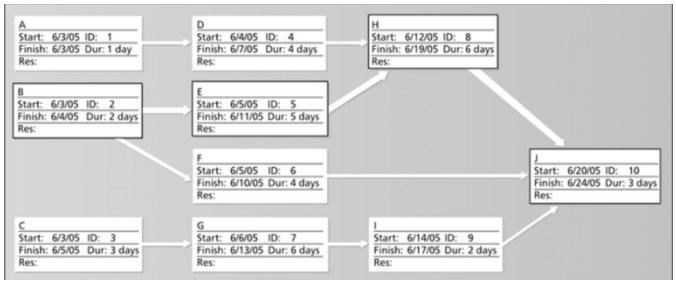


Figure 1-2. Project Management Framework

➢ **Tools and Techniques**

- **Scope Management Tools**
  - ***Project Charter:*** High-level document authorizing the project.
  - ***Scope Statement:*** Defines project objectives, deliverables, and boundaries.
  - ***WBS (Work Breakdown Structure):*** Hierarchical breakdown of project tasks for better management.

- **Time Management Tools**
  - ***Gantt Charts Visual:*** timeline of project tasks and deadlines.
  - ***Network Diagrams Graphical:*** representation of task dependencies.



bolded tasks are on the critical path

  - ***Critical Path Analysis:*** Identifies the longest sequence of dependent tasks to determine project duration.
  - ***Critical Chain Scheduling:*** Focuses on resource constraints and buffers to optimize schedules.

- **<u>Cost Estimate Tools</u>**

  - ***Cost Estimates:*** Forecasts project expenses based on resources and tasks.
  - ***Earned Value Management (EVM):*** Measures project performance by comparing planned vs. actual progress and costs.

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PROJECT MANAGEMENT PROCESS GROUPS

PROCESS GROUPS	INITIATING	PLANNING	EXECUTING	MONITOR & CONTROL	CLOSING
Project Integration Management	• Develop Project Charter	• Develop Project Management Plan	• Direct & Manage Project Work	• Monitor & Control Project Work • Perform Integrated Change Control	
Project Scope Management		• Plan Scope Management • Collect Requirement • Define Scopes • Create WBS		• Validate Scope • Control Scope	
Project Time Management		• Plan Schedule Management • Define Activities • Sequence Activities • Estimate Activity Resources • Estimate Activity Duration • Develop Schedule		• Control Schedule	
Project Cost Management		• Plan Cost Management • Estimate Costs • Determine Budget		• Control Costs	
Project Quality Management		• Plan Quality Management	• Perform Quality Assurance	• Control Quality	
Project HR Management		• Plan HR Management	• Acquire Project Team • Develop Project Team • Manage Project Team		
Project Communication Management		• Plan Communication Management	• Manage Communications	• Control Communications	
Project Risk Management		• Plan Risk Management • Identify Risks • Plan Qualitative Risk Analysis • Plan Quantitative Risk Analysis • Plan Risk Responses		• Control Risks	
Project Procurement Management		• Plan Procurement Management	• Conduct Procurement	• Control Procurement	• Close Procurement
Project Stakeholder Management	• Identify Stakeholders	• Plan Stakeholders Management	• Manage Stakeholders Engagement	• Manage Stakeholders Engagement	